EXPLORATORY DATA ANALYSIS PROJECT REPORT

BIAS AI IN RESUME SCREENING

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Abstract

AI-powered hiring tools can streamline application reading but the results must be examined due to potential disparities arising from past data, proxy variables, model architecture, and human-AI processes. Fair Hire Audit Lab is an educational tool that combines two types of evidence:
(1) a group-disparity audit of standard classifiers learned on the Adult Census Income dataset where class labels are based on whether income exceeds $50K as a proxy target for socioeconomic status;
(2) a paired synthetic-resume test comparing recorded outputs from Gemini and ChatGPT on identical inputs, with identity-controlled changes.
The Adult Census file of 48,842 rows x 15 cols underwent text trimming, question mark replacement with na, row deletion, precise duplicates, and subsampling to yield a training set of 45,175 rows. Logistic Regression and Random Forest models were fit on an 80/20 stratified sample (random seed 42). Accuracy and ROC-AUC

on the test data were 84.47% and 0.9039, respectively, for Logistic Regression and 86.18% and 0.9169, respectively, for Random Forest. Race and gender were held out of the predictor features, yet the trained models showed group differences for the predictor candidate with the largest predictor importance: gender. The pair-wise differences in predicted-positive rates were 17.512 percentage points for the logistic model, 17.288 points for the random forest. For the validated paired LLM, 96 observations were drawn from records R2-5, 48 for each model, and there were zero within-pair decision switches, zero non-zero score diffs, but disagreement for the Profile P3 during Round 3, Gemini derived 72 and shortlister returned SHORTLIST, chatgpt 55 and returned REJECT. This hypothetical project is designed to be broadly interpretable and illustrative. The census target is not a hiring outcome, and the absence of a difference in just two sets of limited paired tests cannot be taken as evidence of fair or unfair models. The project offers an interactive dashboard for investigation, model comparison, pair-wise output examination, and thoughtful interpretation of fairness metrics.

Keywords :

AI-assisted recruitment, algorithmic fairness, resume screening, demographic disparity, counterfactual testing, machine learning audit, explainable decision support

Introduction

Candidates don't win or lose recruiting decisions based only on the skills tested or the job required. They win based on who else applies. In high impact hiring decisions, software is increasingly used to order applications, parse resumes, recognize skills, and create shortlists. Machine-learning models can process applications swiftly and uniformly, but pace and predictive accuracy do not necessarily mean a model is equitable. When trained on historical data, a model can pick up on artificial patterns that have for generations favored some people over others - in schools, at work, and in the job market.

The term AI-assisted recruitment includes a wide range of technically disparate systems: parsing résumés, ranking applicants, assessing responses, analyzing interviews, and even general-purpose large language models (LLMs) prompted to assess candidates. Each of these systems can fail in various ways: a parser might miss a qualification; a ranker might reflect discriminatory patterns from labels in training data; a generative model might respond to a candidate's name or the like; a human reader might simply defer to model output. An audit therefore needs to be explicit about what system is assessed, what metric is used, what comparator is applied, and what the results mean.

Fairness cannot be expressed as a single number. Demographic parity looks at positive rates for a set of groups. Equal opportunity looks at true-positive rates for cases deemed qualified on a reference label. Measures of predictive performance, such as accuracy or ROC-AUC, address a different question: how accurately does the model predict its target? A model can accurately

predict its target, general, but still have unequal group-level positive rates; and having a measured disparity is a reason to look further, not an indication of a specific causal story or illegal discrimination.

The project shows two parallel audit tracks. The first uses the Adult Census Income data set to train two basic classifiers of an income-related target, then compares model outputs by subgroup. This is not a resume data set, and income greater than $50K is not a hiring label. It is used as a proxy task to show calculation and interpretability of group metrics. The second uses a saved set of paired LLM outputs for synthetic candidate profiles; within each pair, the prompt-tuned intent changes an identity-associated indicator while keeping qualifications constant, then the score and decision are compared.

The last app, Fair Hire Audit Lab, wraps up these analyses in a Streamlit app. Its pages are a final dashboard, dataset exploration, income-proxy model audit, paired LLM results, a resume input workflow, a counterfactual prompt lab, and exports. The resume input workflow can show a local keyword-coverage baseline, but live Gemini screening is optional since it requires external API access and is not needed to interpret the saved audit results. The final project is thus an audit and teaching demo rather than a production hiring tool.

1.1 Problem statement

Candidate filtering with the help of AI might produce varying outcomes within a single group or between providers of a model, despite the fact that each provides a single accuracy figure. An open and straightforward process should be employed to assess data quality, benchmark models, quantify the gaps across groups, and compare pairs of LLM outputs.

1.2 Aim and objectives

- Construct a dashboard recording data prep and investigating the Adult Census dataset.
- Train and compare Logistic Regression and Random Forest models for the income-proxy classification.
- Evaluate the predicted-positive rates of the measure group and the obtained fairness-measures by gender and by race.
- Summarize all recorded paired LLM results by difference in score and amount of decision changes.
- This is a controlled demonstration of prompt creation and resume screening for class room use.

- State the limitations of proxy labels, sample size, model coverage, and causal inference.

1.3 Scope and contributions

This contribution is an integrated, reproducible classroom audit interface, which displays the two evidence streams alongside each other and keeps the differences between them intact. This contribution does not aspire to reproduce any commercial hiring system, to perform legal discrimination, or to demonstrate a model's overall fairness. The saved paired LLM analysis includes only Rounds 2-5; the earlier Round 1 records were not included in the validated file due to inconsistent record in those records.

2. Related Work and Literature Survey

Research on algorithmic hiring has detailed risks along the entire pipeline including in data collection, defining the target, constructing features, training models, deploying systems, and for human users. (Raghavan et al., 2020; Bogen & Rieke, 2018; Dadaboyev et al., 2025). Raghavan et al. (2020) investigated the industry claims about algorithmic hiring and noted that vendor claims and technical report must be interpreted with caution, and Bogen and Rieke (2018) provided a range of risks from the process of search and screening through assessment along automated hiring workflows. Together this research underpins audits that probe more than the purported efficiency of a system.

They inform the framework established by a wide body of algorithmic fairness literature. Sources of bias and mitigation approaches in machine learning is outlined by Mehrabi et al. (2021), while considerations of harms throughout the machine-learning lifecycle are detailed by Suresh and Guttag (2021). Selbst et al. (2019) point out that an inability of the fairness intervention to achieve the desired results can be attributed to separation of social systems from the technical system, while Barocas, Hardt, and Narayanan (2023) systematically cover the concepts, metrics and trade-offs of fairness. These references justify that the metrics are reported together with the information about the context and limitation of the target label. (Mehrabi et al., 2021; Suresh & Guttag, 2021; Selbst et al., 2019; Barocas et al., 2023).

Counterfactual and audit-style testing provides a way to isolate one factor. In a paired design, the same latent profile is tested in two versions that differ only in one attribute. This reduces some confounding because educational background, experience, and skill level are held constant, and the profile is the same in each version. But the validity of the conclusion still relies on whether the prompt preserves all other content, whether the identity manipulation is plausible, whether the model is deterministic, and whether the sample has enough profiles and enough repeated runs of each pair. One pair is a diagnostic case, not an estimate of the entire population. (An et al., 2025; Wilson & Caliskan, 2024).

Studies of language technologies have found social signals in embeddings, classification, and generation. Buolamwini and Gebru (2018) found intersectional performance gaps in commercial

gender classifiers. While face analysis is a distinct kind of task from screening resumes for potential hiring candidates, that study is one of the most well-known examples of disaggregation of results by group, rather than just looking at overall performance. Binns (2018) theorizes about the meaning and feasibility of fairness in machine learning, and explains that the conflicting nature of different fairness metrics is part of what makes it necessary to measure, rather than simply assume from system design intentions. Kearns and Roth (2019) give some of the motivation for formalizing and measuring fairness (in contrast to an a priori assumption that systems should be fair based on their design) (Buolamwini & Gebru, 2018; Binns, 2018; Kearns & Roth, 2019).

Slightly more directly, the material cited in the Review 2 literature survey discusses research on AI recruitment, gender and racial bias, cross-model differences, anonymisation, and human-AI decision-making. The project report's source material includes An et al. (2025) on gender and racial bias in automated resume evaluation; Wilson and Caliskan (2024) on gender, race, and intersectional bias in resume screening using language-model retrieval; and the 2024 piece "Computer says 'no'" on systemic bias in ChatGPT using an audit methodology. All of these produce relevant precedent for controlled audits. Their sample sizes and findings are not otherwise-generated results and are not generated by Fair Hire.

The literature also cautions that just because explicit protected attributes are removed does not mean this removes their effect entirely. Even though other features may be correlated with protected groups, they can act as a proxy. Our census audit omits

race and gender from our model feature matrix, however we do include it with for post-hoc group analysis, which enables us to investigate groups while not putting those columns directly into the classifier, knowing that occupation, education, hours worked and other features can encode this information (Barocas & Selbst, 2016; Mehrabi et al., 2021).

Another issue is the link between a model's output and the decision of a person or company. An automated score can be used as a decision and not only a suggestion. The decision of a human judge can be influenced if the score is perceived as an unbiased suggestion or a human may not have the time to argue the proposed actions. Therefore, in a fair-hiring process, a score needs to be for governance, to have a documentation, a review process, and an appeal process, in addition to a metric for the model. The Fair Hire website states its scores to be only advisory, and therefore provides no automatic rejection. (Lacroux & Martin-Lacroux, 2022; Raghavan et al., 2020).

2.1 Research gap

The literature survey provided in the supplied documentation highlights several issues relevant to a classroom-scale prototype: few standardized benchmark criteria across systems; research that tackles only one protected attribute or one job (role) scenario; lack of information about the process that led to a given result; need to clarify the difference between group disparities in the prediction of a proxy and real hiring. This work addresses these points by recording its data-processing decisions, providing two baseline

models, preserving paired output records, and clearly separated, two audit tracks. Its small paired sample is a limitation, not a replacement, for large scale validation.

2.2 Project contribution relative to prior approaches

Dimension	Common audit approach	FairHire implementation	Limitation
Dataset audit	Measure model performance and group rates	Adult Census proxy classifier with LR and RF	Not hiring data; proxy target only
Counterfactual test	Compare matched profiles under controlled changes	96 recorded paired observations across R2–R5	Limited prompts, profiles, runs and models
Metrics	Accuracy and group fairness indicators	Accuracy, balanced accuracy, F1, ROC-AUC, rate gaps and ratios	Metrics can conflict; group sizes differ
Reproducibility	Document data, model, and outputs	Saved CSV, fixed split seed, dashboard and export	External API workflow can fail or change
Responsible interpretation	Discuss scope and possible harms	Explicit proxy-label and non-causality warnings	Not a legal compliance assessment

3. Proposed Methodology

The pipeline was separated into two different audit pipelines via a single application.

Pipeline A was used for exploratory data analysis and model auditing of the Adult Census Income dataset.

Pipeline B is used for validation and summarization of the saved paired LLM output file. The pipelines were not combined into a single hiring-bias score as their labels, units of analysis, and questions were different.

3.1 System architecture

Pipeline stage	Input / operation	Output
A1. Data input	Adult Census CSV: 48,842 rows, 15 columns	Raw dataset and data-quality summary
A2. Cleaning	Trim strings; convert '?' to missing; drop incomplete and duplicate rows	45,175 cleaned unique rows
A3. Model training	Stratified 80/20 split; random_state=42; LR and RF; race/gender excluded from features	Predictions on 9,035-row held-out set
A4. Fairness audit	Group predicted-positive rates, gaps, ratios and performance metrics	Model and group comparison tables
B1. Paired output input	Validated CSV from Rounds 2–5, 96 observations	Recorded A/B decisions, scores and metadata
B2. Pair analysis	Compute A–B score differences and decision flips	Within-pair summary and cross-model comparisons
UI / reporting	Streamlit dashboard, charts, tables and export functions	Inspectable results and discussion

Figure 1. Architecture summarized as data input → cleaning and validation → model training / paired-output validation → metric calculation → dashboard and exports. The two analytical pipelines remain separate until the dashboard presents their results side by side.

3.2 Adult Census data preparation

The CSV file provided had 48,842 rows and 15 columns. Trimming of the column names and string values was performed. Missing values include mark-up '?' question-marks, as well as null String values were blank rows were dropped resulting in 45,175 clean unique records. The income feature was made into a binary target value, with positive meaning income above $50K. Race and gender were left as a separate attribute (for post-hoc analysis), while not part of the predictor feature matrix, precluding direct access to either of those two columns, but not precluding information in those two features be encoded for other attribute columns. Splitting of training and testing Data was split into training and testing using an 80/20 stratified split (random_state=42). Stratification ensures that the proportions of the target-class are similar in the split. Pipelines of preprocessing and modeling were applied to categorical and numerical fields. The held out test set consisted of 9035 rows. As the outcome is income we interpret any group-rate result as an income -prediction disparity rather than a hiring or shortlisting disparity.

3.3 Classifier models

Logistic Regression, which models the log-odds of the positive class as a linear combination of the features after transformations. It provides a useful point of comparison for a more flexible model. Random Forest makes predictions by combining predictions from

many decision trees, and can model nonlinearities and interactions. Comparing the two gives the project a way to verify whether high-level performance differences and group-rate patterns appear for a single family of models, or are present for both. Model hyperparameters are held at the documented baseline setting, rather than being tuned to optimize a fairness metric.

3.4 Fairness and performance measures

Accuracy: number of correct predictions over total. Balanced accuracy: averages recall across classes and is useful when class proportions are unbalanced. F1: harmonic mean of precision and recall for the positive class. ROC-AUC: evaluates the ability of the model to rank data points across thresholds. These metrics are described for the predictive performance against the income proxy. For a group g the predicted-positive rate (the percentage of the group g that is predicted positive) is the percentage of group members labeled positive. The selection-rate gap (the difference in those percentages across groups) is maximum predicted-positive rate minus minimum predicted-positive rate (measured in percentage points). The minimum-to-maximum ratio (minimum predicted-positive rate divided by maximum predicted-positive rate) is sometimes used as a screening indicator for further review (flag if less than 0.80). It is not a legal conclusion and should not be used as such without being given its context. Equal opportunity is the rate of acceptance (true positive rate) among those who were marked positive in the reference label.

3.5 Paired LLM audit

The validated paired file contains 96 observations from Rounds 2–5: 48 Gemini observations and 48 ChatGPT/GPT-6 observations. Each row records the pair identifier, profile, varied attribute, A/B values, decisions, scores, rationales, run identifier, model and prompt condition. The score difference is Candidate A's score minus Candidate B's score. A decision flip occurs when the recorded A and B decisions differ. The app summarizes these records rather than generating new LLM outputs automatically.

A paired audit is informative when the candidate qualifications are held constant and only the identity-associated signal changes. In practice, the test should preserve the exact resume text except for the manipulated field, use a consistent job description and scoring rubric, save the raw outputs, and repeat across profiles and model versions. This project reports only the validated saved records. Round 1 was not merged because the previously available Round 1 records conflicted; excluding inconsistent records preserves a clearer audit trail.

3.6 Resume workflow and application design

The Resume input workflow of the Streamlit app requires synthetic or anonymized PDF, DOCX and TXT resumes, a target job description and list of needed skills. The local keyword-coverage baseline is a simple test whether the needed skills are found in the extracted resume text. This is a visible demo (not an LLM fairness measure). The Gemini path in this live version is optional, relying

on API access, quota, network latency and model availability. An counterfactual workflow allows the submission of two versions of a resume with one attribute related to an identity changed. The app states these results are indicative and not for automatic rejections.

3.7 Ethical safeguards

- Use demo or anonymized resumes for demonstrations. Do not upload applicant information with PHI.
- Maintain the model scores as advisories and have a human verify the evidence and missing qualifications.
- Do not jump to a conclusion that a difference indicates intention or unlawful discrimination.
- Note the model name, prompt, date run, output, and failure for reproducibility.
- Never use the Adult Census income goal to measure hiring success or candidate quality.

3.8 Implementation details and data flow

The app is served as a local Streamlit dashboard. The input files are selected relative to the python application file so that the app is not affected by the working directory of the terminal. The Adult Census file is loaded and column-named, text values trimmed and missing-value indicators normalized prior to cleansing. The resulting dataframe is fed into the exploratory views and into the fitting

model as the data source. The paired audit file is loaded on its own and checked for expected schema prior to calculating summary statistics. Listing the separate input paths independently prevents the accidental shuffling of census records with resume level observations.

The user interface is paged to facilitate in navigating the analysis. The dashboard page summarizes results and the interpretation boundary. The EDA page view quality of data and the distributions of the data. The model-audit page view the model performance and group details by demographic features. The paired LLM page view the saved pairs-level outputs. The resume input page accepts an anonymized or synthetic document and a position description, on the local baseline page, search for the required-skill terms in the document. The prompt lab generates a limited prompt for execution by an external LLM, and the export page expose selected summaries for inspection. With this componentized design, one can observe the completed analysis even without the optional external API.

Baseline for local keyword is as minimal as possible: it only looks for our specific strings in the text extracted from the document, it doesn't know about synonyms, similar skills, context, and may sometimes hit on a skill that wasn't used (say, it finds 'structured query language' but we wanted 'sql'). And it has its edge cases: a keyword in a sentence that says "not" on the line before may get a hit, or a skill that isn't relevant to a large section of the doc might do the same. So, it's definitely a good baseline for the skill that we can read docs in (and hit on some skills), but not an ML hiring score. The 'did we encounter a warning on the dashboard?' is a very important part of our implementation!

3.9 Reproducibility and audit trail

Reproducibility requires noting the dataset version, cleaning rules, split specification, model family, and outcome definition. The Adult Census analysis runs a stratified 80/20 split with random_state=42 so that, assuming the same cleaned data and software, the split can be generated with new runs. Results are likely to change with different source data, different library versions, differences in any preprocessing, or even a different version of scikit-learn with the same hyperparameters. For a more formal extension, the application should produce a run manifest that recorded the file hash, timestamp, package versions, random seed, model parameters, and all relevant metrics.
For paired LLM testing, a record to reproduce should have model, run date, prompt template, job description, profile ID, identity field to vary, candidate A value, candidate B value, raw response, extracted score and decision, API error. The validated CSV has pair ID, model, run ID, score, decision, rationale, prompt condition, so can summarise the existing observations. Note that the saved Records cover R2-5 so report has that precise scope. Unlike the default of silently resolving/merging conflicting Round 1 values (for easy batch submission) it's preferable to merely grow the sample without internal inconsistencies.

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4. Experimental Analysis, Comparison and Results Discussion

4.1 Classifier performance

Both classifiers were also scored on the same held-out test set. Random Forest exhibited superior scores on the metrics used to report predictive accuracy, but neither metric indicates whether the model is fair.

Metric	Logistic Regression	Random Forest
Accuracy	84.47%	86.18%
Balanced accuracy	75.91%	77.39%
F1 score	65.30%	68.26%
ROC-AUC	0.9039	0.9169

Table 2. Held-out performance for the Adult Census income-proxy task. These figures describe prediction of income above $50K and should not be presented as resume-screening accuracy.

4.2 Gender group rates

Model	Male predicted-positive rate	Female predicted-positive rate	Gap	Min/max ratio
Logistic Regression	25.70% (n=6,069)	8.19% (n=2,966)	17.512 pp	0.319
Random Forest	24.44% (n=6,069)	7.15% (n=2,966)	17.288 pp	0.293

Table 3. Gender disaggregation of predicted-positive rates in the income-proxy task. The male and female rates differ substantially in both models. Ratios below 0.80 are flags for investigation under a common heuristic, not standalone evidence of unlawful discrimination. The target and dataset context must remain explicit.

4.3 Race group rates

Race group	Logistic Regression	Random Forest
Asian-Pac-Islander	28.30% (n=265)	26.79% (n=265)
White	21.14% (n=7,748)	19.90% (n=7,748)
Amer-Indian-Eskimo	9.59% (n=73)	8.22% (n=73)
Black	9.28% (n=873)	8.36% (n=873)
Other	2.63% (n=76)	3.95% (n=76)
Maximum–minimum gap	25.67 pp	22.85 pp
Minimum/maximum ratio	0.093	0.147

Table 4. Race-group predicted-positive rates. Some categories have relatively small sample sizes, especially Amer-Indian-Eskimo and Other. Rates for smaller groups can be unstable and should be accompanied by counts. The differences indicate model behavior on the proxy task; they do not identify the cause of the disparities and do not represent actual candidate selection rates.

4.4 Paired LLM outputs

Measure	Observed result	Interpretation
Validated paired observations	96 (48 per model)	Rounds 2–5 only
Within-pair decision changes	0	No decision flips detected in these recorded pairs
Within-pair nonzero score differences	0	No A/B score differences detected in these recorded pairs
Largest cross-model disagreement	P3, Round 3: Gemini 72 / SHORTLIST; ChatGPT 55 / REJECT	17-point difference between models, not a within-pair demographic effect

Table 5. Summary of the recorded paired LLM audit. Zero within-pair differences mean that this particular test did not detect an

effect under its prompts, profiles, attributes, and recorded runs. It does not prove that Gemini, ChatGPT, or other models are generally unbiased. The cross-model disagreement demonstrates that model choice can affect an evaluation even when within-model A/B comparisons are identical.

4.5 Comparison with existing approaches

Related work has used large resume corpora, randomized signals of identity, semi-simulated recruitment experience, and studies of language model comparison. The literature sample contains publications that present gender- or race-related biases in their data under their particular conditions. Fair Hire differs from the size and motivation of those works. Fair Hire is a reproducible classroom prototype based on a small validated paired-output file and a second conventional ML proxy audit - not a restudy of those papers' large-scale experiments.

Aspect	Published audit studies (as described in reviewed sources)	FairHire results
Scale	Some studies use tens or hundreds of thousands of resume/job combinations	96 paired observations in the validated R2–R5 file
Outcome	Often simulated screening scores, rankings, or callbacks	Recorded fit scores and decisions; separate income-proxy predictions
Identity treatment	Randomized or controlled demographic signals	Paired A/B identity-associated changes in saved records
Result	External studies report effects under their own conditions	No within-pair differences in validated records; one cross-model disagreement
Inference	Bounded to their prompts, systems and samples	Bounded to the saved records and proxy classification task

Table 6. Qualitative comparison. External studies' results are not counted as FairHire results. A direct comparison of effect sizes would require matching prompts, datasets, models, outcome definitions, and statistical procedures.

4.6 Discussion

The findings explain the reason for assessing accuracy and fairness independently. While RF's accuracy and ROC-AUC is higher than LR, both classifiers reveal large group disparities in the predicted-positive rate, which is unaffected when race and gender are not in the input space, likely reflecting other features' correlated information. This finding prompts an analysis of data origin, feature set, target definition, thresholds, and mitigation strategy.

The paired LLM track offers a different form of evidence. No differences in scores or decisions were observed between the 96 pairs that were recorded, so the current test did not find identity sensitivity in this case. This should be reported in a straightforward manner, rather than exaggerated to imply general fairness. To conduct a more extensive audit, we would need more profiles, multiple repetitions of the same test, different model versions, more job families, flipped identity roles, and a preregistered analysis plan. Model output may also fluctuate over time due to API updates or nondeterminism.

These two tracks must not be conflated. In the Adult Census, the task predicts income over $50K; in the paired LLM track, the task tracks synthetic resume evaluations. They differ in data, target, unit of analysis, and interpretation. In the first, we record group

inequality on a socioeconomic proxy classification task; in the second, we record no within-pair difference on a limited controlled experiment. Neither tests how a real applicant is treated by a deployed hiring platform.
The 0.80 ratio heuristic is fine as a prompt for further investigation, but can't tell us if a ratio is legal or fair by itself. Group definitions, sample sizes, job relevance, label quality, decision thresholds, and uncertainty all are relevant. To understand the stability of rates with small group counts, the project dashboard should provide these counts as well as rates, and some indication of the interpretation boundary.

4.7 Limitations

- The Adult Census data is not a resume or recruitment data - income > 50K is just a proxy label.
- The pairwise LLM sample includes 96 observations from Rounds 2-5 and excludes some profiles, jobs, attributes, prompts, and model versions.
- Round 1 was omitted as there were discrepancies in the records available; therefore the final paired analysis is not an exhaustive record of all attempted rounds.
- Race and gender do not directly appear as features in the model, but depending on the features included, proxy information may be encoded.
- Difference in group level rates are merely descriptive and cannot demonstrate the presence of a causal mechanism and of unlawful discrimination.

- Live Gemini screening requires access to an external API, so will not be performed for saved-results analysis.
- The local keyword baseline matches text against a list of skills and is not equivalent to semantic or human assessment.

4.8 Interpreting the fairness indicators

A group selection-rate gap compares the maximum and minimum positive prediction rates across the groups under study. For example, the maximum-to-minimum group prediction rate gap gives the maximum predicted positive rate as a percentage of the minimum; the measure is intuitive because it compares percentages but is very sensitive to which groups are chosen. The ratio of the minimum to the maximum gives the lowest rate relative to the highest; a ratio close to 1 indicates the rates for the two groups are similar, while a low ratio indicates the two groups differ and may merit further study. Neither measure reveals the mechanism causing the disparity, and neither measure contains information on the uncertainty of the measures unless additional analysis such as confidence intervals or statistical testing is performed.
Equal opportunity and equalized-odds-style metrics include the reference label. A true-positive rate is the proportion of positive instances (according to the reference label) that are labeled as positive; a false-positive rate is the proportion of negative instances that are labeled positive. These metrics can provide insights into behavior at different error rates that a comparison of selection rates might miss; the latter is an artifact of the problem it describes, whereas the interpretation of the former depends on the label used. In this project, our label was income over $50K; in this context, "qualified" in a proxy TPR calculation indicates positive under the

income label, not qualified for a certain vacancy. We refer to the target label as proxy target.
The fairness metrics need not all go in the same direction. Equalizing selection rates could worsen or could be equalize error rates if group base rates differ, and raising the cutoff to catch more positives in one group may increase false positives or false negatives in that group. An audit in good faith should report many different numbers, describe what is being optimized and the threshold for action, and account for trade-offs, rather than produce one single supposedly fair number. This dashboard is diagnostic; it does not provide a mitigation technique or certify a system for a legal standard.

4.9 Practical implications for a human-reviewed workflow

For a live organization, the initial requirement is to establish the job-related requirements, before examining the models' outputs. For each requirement, ensure it is justifiable by the role, applied consistently, and mapped to material found in the resume. A reviewer should be able to trace which parts of the input caused the system to assign the score, rectify an extraction error and log when the system is unsure. Absence of information should be distinguished from evidence of absence that a candidate has the requirement. These safeguards limit the potential for an automated score to be treated as an infallible decision.
A second control is "post-deployment" regular monitoring. The set of candidate pools, or hiring needs, may shift and historical logs

may become stale, and an audit done at one time may no longer be valid. Monitoring should measure group counts, group rates, error patterns, and model drift across groups for all relevant roles and time periods. When a material difference is identified, it should further be evaluated and addressed by examining data quality, proxy features, thresholds, and the workflow; and tested for effects on both performance and fairness, and checked with the relevant legal, HR, and stakeholder communities.

The prototype depicts some aspects of this workflow-data-quality checks, group metrics, saved model outputs, human-review framing-but lacks the organizational control structure for deployment. It is not integrated with applicant-tracking software; it does not verify employment-law compliance; it does not complete a representative field test. Its purpose is to be instructive: to show how to identify quantifiable audit questions, how to audit outputs, and how not to conflate predictive accuracy and fairness.

4.10 Extended comparison with prior work

The conventional-model work that most resembles this project is work that conceptualizes fairness as a lifecycle issue: decisions about the dataset and label structure shape what the model learns, then group-level analysis must be done post-training to detect potential disparate impacts. It complements the evaluation of commercial hiring products that analyzes claims made by vendors and how much vendor validation is disclosed. The Adult Census proxy task provides a publicly available, replicable demonstration, but it does not provide information about the behavior of a commercial resume screening tool due to differences in the task,

features, and population.
Pairwise LLM approach is structurally more similar to counterfactual language-model screening audits. With all other signals fixed save one, pairwise identity comparison can be meaningfully aggregated at the pair level. But statistical power with this design is a function of the number and variety of pairs, run repetitions, inclusion of all pairs of models in each run, and careful randomization and control of prompt formulations. This 96-row file is only sufficient for pair-difference averaging and cross-model comparison, not precision estimation of small effects in the population of all possible occupations, nor extrapolation to all possible occupation models. Zero pair-differences recorded should thus be taken as true negative detections for these conditions.
Both tracks jointly give a learning-focused contrast between the difference in predictions across aggregate groups and the sensitivity of predictions within a pair to changes in a protected feature. Aggregate rates indicate that a model's predictions may differ by group, despite an absence of protected fields in the training data. Paired tests can help spot one controlled identity-based change, but miss effects beyond the chosen prompts and profiles.

5. Conclusion and Future Work

Fair Hire Audit Lab combines: exploratory data analysis, two baseline classifiers, group-level proxy metrics, and a combined paired LLM results dashboard in a single educational tool. On the Adult Census income-proxy task, Random Forest beat Logistic Regression on the reported summary metrics but the two models had strikingly large gender- and race-based group disparities in

predicted-positive rate (PPR). In the valid paired LLM records from Rounds 2-5, no decision flips between pairs and no differences in the difference of scores across both models. There was a crossed model disagreement for Profile P3 in Round 3. The results highlight the need to disentangle model performance, group disparity, within-pair model-sensitive flips, and cross-model compositional disagreement.

The project does not demonstrate that a genuine hiring system is discriminatory or not. The Adult Census results apply to a socioeconomic proxy target, and the paired LLM results apply to a small set of recorded synthetic-profile tests. The dashboard's primary benefit is exposing assumptions, metrics, and limitations for interrogation and reproducibility.

Future work should extend the paired audit with more synthetic profiles, occupations, attributes, runs, reversed assignment of identities, multiple model versions. It should report uncertainty intervals and follow a preregistered protocol. If access and governance conditions allow, future work could compare outputs to human-reviewed job-relevant criteria and test a real hiring dataset with appropriate consent and safeguards. Other work could provide fairness-aware training, threshold sensitivity analysis, model prediction explanations, audit logging, and a structured human-review pipeline. Any mitigation should be assessed on both the accuracy and potential tradeoffs.

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